Ultimate LaTeX Cheat Sheet: Math, Graphs, and Vectors

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1. Basic Math Symbols

- Superscript: x^2, y^n
- Subscript: x_1, a_{ij}
- Fraction: $\frac{3}{4}$
- Square root: $\sqrt{16}$, $\sqrt[3]{8}$
- Summation: $\sum_{i=1}^{5} i = 15$
- Integral: $\int_0^1 x^2 dx = \frac{1}{3}$
- Greek letters: $\alpha, \beta, \gamma, \delta, \theta, \lambda, \mu, \pi$
- Inequalities: $3 < 5, 7 \geq 4, 5 \neq 2$

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2. Vectors and Operations

- Vector notation: $\vec{v} = (2, 3)$
- Vector difference: $\vec{AB} = B - A$
- Magnitude: $\|\vec{v}\| = \sqrt{2^2 + 3^2} = \sqrt{13}$
- Unit vector: $\hat{v} = \frac{\vec{v}}{\|\vec{v}\|} = \left(\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}}\right)$

Example:

$$A = (1, 1), \quad B = (4, 3) \Rightarrow \vec{AB} = (3, 2)$$

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3. TikZ / pgfplots Basics

```
\begin{tikzpicture}
\begin{axis}[axis lines=middle, xlabel=$x$, ylabel=$y$]
\addplot[domain=-2:2, blue]{x^2};
\end{axis}
\end{tikzpicture}
```

- Plots $y = x^2$ in blue. - Axes cross at origin.

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4. Plotting Lines

```
\addplot[color=red, thick]{2*x + 1};
\addplot[color=green, thick, dashed]{-x + 2};
```

- Plots two lines: $y = 2x + 1$ (red) and $y = -x + 2$ (green dashed).

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5. Plotting Points and Labels

```
\addplot[only marks, mark=*] coordinates {(1,1) (2,3)};
\node at (axis cs:1,1)[above right]{$A$};
\node at (axis cs:2,3)[above right]{$B$};
```

- Points at $(1, 1)$ and $(2, 3)$ labeled A and B.

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6. Drawing Vectors (Arrows)

```
\draw[->, thick, blue] (axis cs:1,1) -- (axis cs:2,3);
```

- Draws vector $\vec{AB}$ from point A to B in blue.

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7. Combined 2D Example

```
\begin{tikzpicture}
\begin{axis}[axis lines=middle, xlabel=$x$, ylabel=$y$]

% Points
\addplot[only marks, mark=*] coordinates {(1,1) (4,2)};

% Labels
\node at (axis cs:1,1)[above right]{$A$};
\node at (axis cs:4,2)[above right]{$B$};

% Vector
\draw[->, thick, blue] (axis cs:1,1) -- (axis cs:4,2);

% Line
\addplot[color=red, thick]{0.25*x + 0.75};

\end{axis}
\end{tikzpicture}
```

- Combines points, labels, vector, and line.

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8. 3D Plots and Planes

```
\begin{tikzpicture}
\begin{axis}[view={60}{30}, xlabel=$x$, ylabel=$y$, zlabel=$z$]
\addplot3[surf, domain=-3:3, domain y=-3:3]{x + y};
\end{axis}
\end{tikzpicture}
```

- Plots plane $z = x + y$ in 3D. - ‘view=6030‘ rotates the axes for better view.

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9. Styling Tips

- Line thickness: thick, ultra thick
- Dashed lines: dashed
- Colors: blue, red, green, black
- Mark styles: *, square, o
- Add legend: `\addlegendentry{$y=x+2$}`

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10. Quick Reference

- Line: $y = mx + c$
- Plane: $z = ax + by + c$
- Point: (x, y)
- Vector: $\vec{v}$
- Vector difference: $\vec{AB} = B - A$

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11. Notes

- Always use math mode for variables: x not x
- Use color, labels, and line style for clarity
- Start small: lines → points → vectors → planes → 3D